Airframe Technology Master

Certification
Lecture 01

TLAR. Airworthiness regulations

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April 2025

AGENDA

- TOP LEVEL AIRCRAFT REQUIREMENTS
- CERTIFICATION REQUIREMENTS
- AIRWORTHINESS IN MILITARY AVIATION
- QUALIFICATION

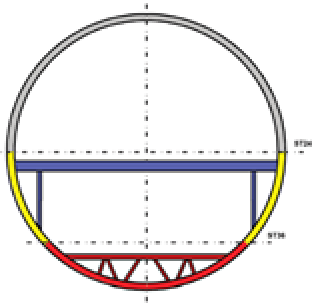
TOP LEVEL AIRCRAFT REQUIREMENTS

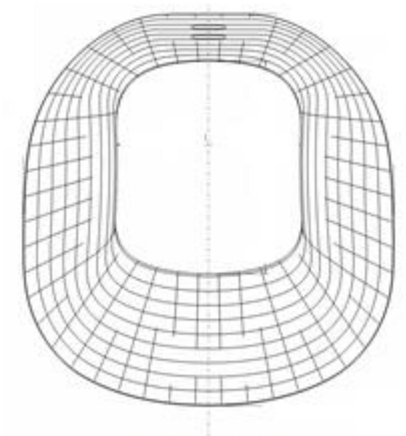
TOP LEVEL AIRCRAFT REQUIREMENTS

A set of Top Level Aircraft Requirements (TLARs) is defined in the beginning of a project.
The TLARs and the general configuration decision are used as input for the design.
TLARs are basically established from market requirements but…not only.

MAX Presurization Cabin pressurization:

Pressurization becomes increasingly necessary at altitudes above 10,000 feet (3,000 m) above sea level to protect crew and passengers



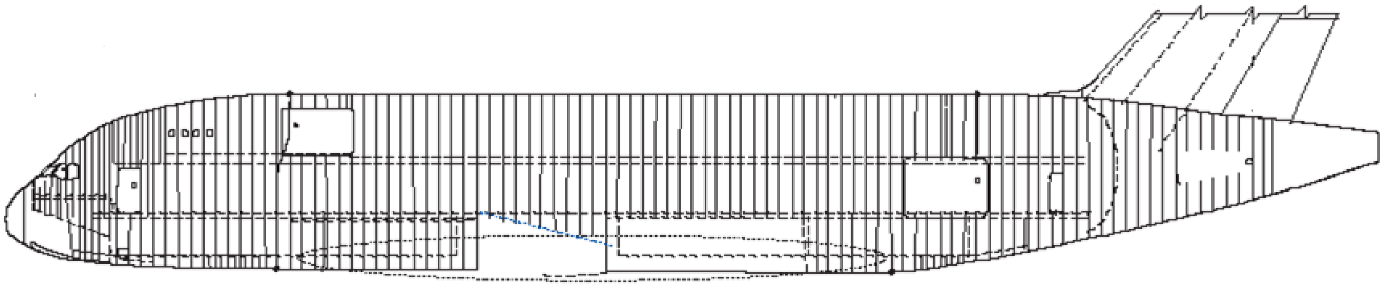


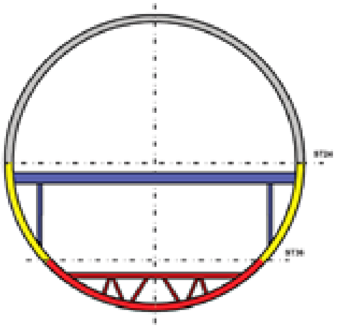
FRONT VIEW

Altitude:

Cabin pressurization:

Pressurization becomes increasingly necessary at altitudes above 10,000 feet (3,000 m) above sea level to protect crew and passengers



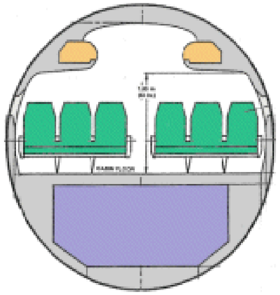


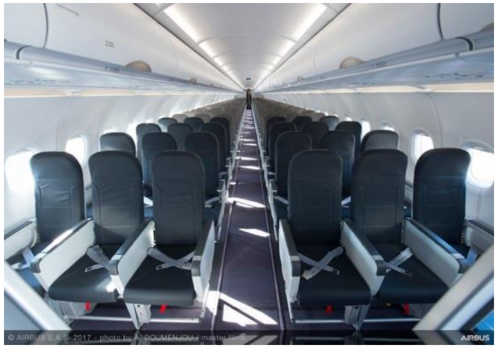
Cross section
Typical pressurised fuselage

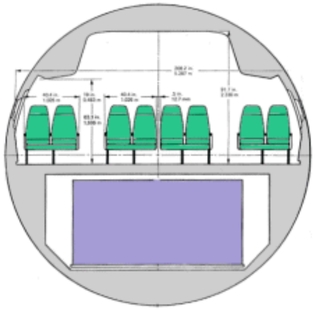
TOP LEVEL AIRCRAFT REQUIREMENTS

Pax Number:

The seat abreast influence the cross section of fuselage to allocate passengers and cargo







Single aisle pax abreast (A320 family)

Twin aisle pax abreast (A330 family)

Specific Military TLAR

Specific Military TLAR

- **Cargo Pay Load**

Specific Military TLAR

- Low Lever Flight

Specific Military TLAR

- **Tactical Take off and Landing**

Sizing Impact :

- LG
- WING
- FUSELAGE

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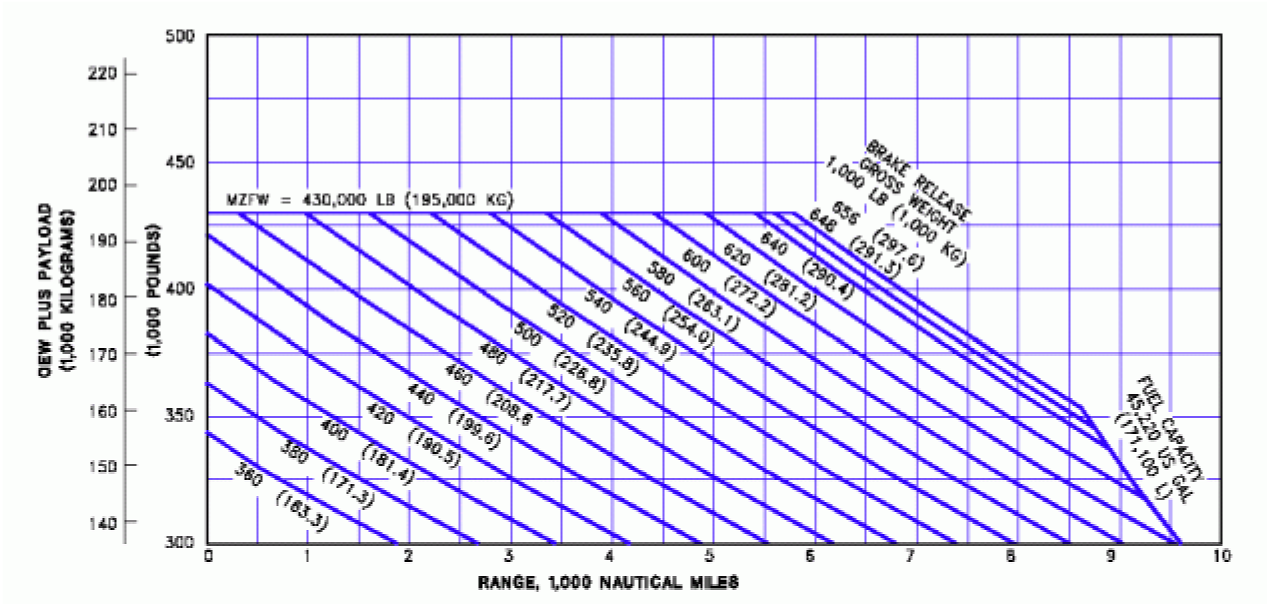
Specific Military TLAR

TLARs

Some TLARs may correlate in a adverse way.



3.2.2 PAYLOAD/RANGE FOR 0.84 MACH CRUISE
MODEL 777-200 (HIGH GROSS WEIGHT AIRPLANE)

Early in the design synthesis phase, trading off is the traditional approach used to understand and evaluate aircraft design sensitivities and margins.

TLAR

The three main parties defining the most important requirements are:

- **Regulator** (i.e the government authorities, EASA, FAA) → **SAFETY →MANDATORY!**
- The operator →Minimum Total Cost of Ownership
- The aircraft manufacturers →Attractive product at low manufacturing cost

REGULATOR

Environmental Friendliness

Regulatory Requirements

Safety

OPERATOR

Easy Maintainability

High Conform

Low Price

High Flexibility

Long Service Life

Minimum Weight

MANUFACTURER

Existing Design, Rules, Principles

Available Manufacturing Technologies

Available Resources

Low Manufacturing Cost

Qualified Materials

TLAR

- The compilation, definition and documentation of requirements is crucial to mitigate risks (delays, cost increase, industrialization problems)

- Precise definition of functions is key

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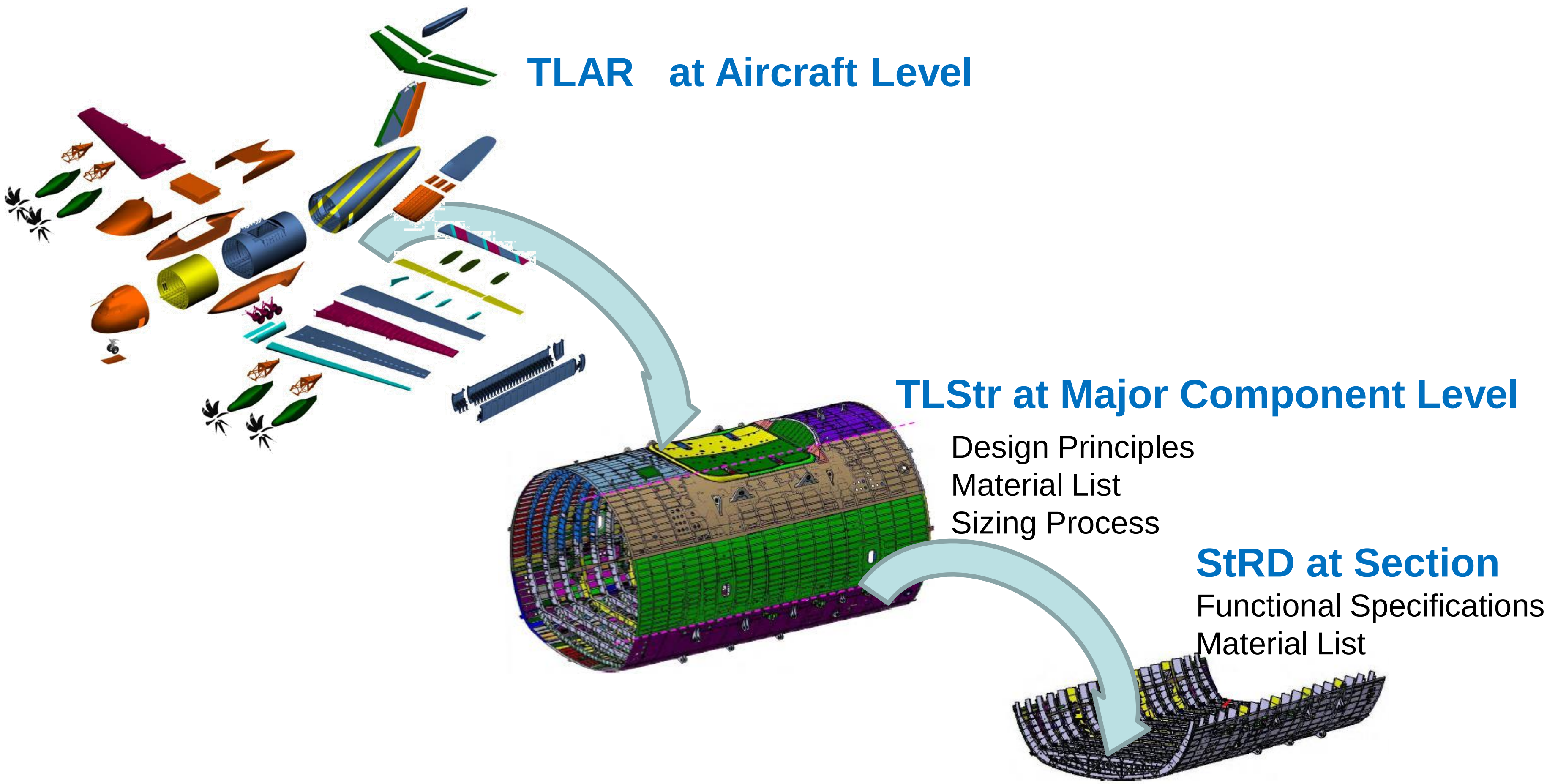
TLAR

These requirements are broken into requirements for Major Components of the A/C, Top Level Structure Requirements Dossier (TLSrRD) e.g for:

- Fuselage
- Wing
- Empennage
- Systems

- And **cascade** to Section Level Requirements (StRD) for structural components e.g. for
- Skins
- Stringers
- Floor structure

TOP LEVEL AIRCRAFT REQUIREMENTS



CERTIFICATION REQUIREMENTS

Certification Requirements - Why?

The certification requirements are part of the top level requirements that an aircraft has to met and are as important as the programme and customer requirements.

This is because the international aviation organization requires that every aircraft is in a condition for a **safe** operation.

During the whole life of an aircraft, from the very beginning of the A/C design to the in service operation, the aircraft has to be compliant with the **aviation safety** and applicable **regulations**

PRIOR TO START … SOME IMPORTANT DEFINITIONS

What is **Airworthiness**?

Is to keep to acceptable levels the probability of an accident or incident as a result of any malfunction, performance or handling of the aircraft.

Airworthiness is the capability to fly in a **safety** way for the A/C, for the transported passengers and goods and for the flown territories

What is **certification**?

Is to stand in writing from an appropriate authority that a reality conforms with the truth.

To certify an aircraft is to obtain a **document** which stand that the aircraft conforms with the regulations and it is suitable for a **safe flight**

Aviation safety – The basics: Who and How?

Aviation safety is a **global public right**:

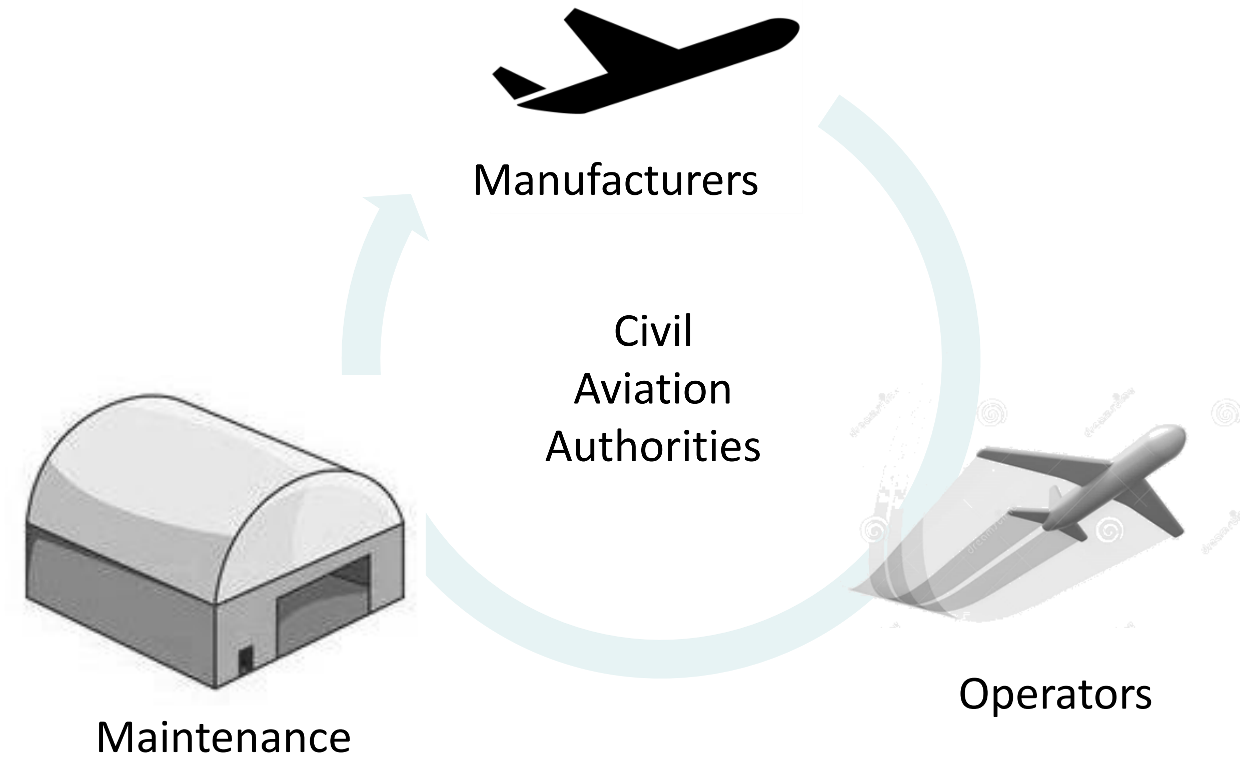
— Public stands for non private, non marketable right. It is ensured by the **law**. National or international legal framework regulate all aspects and all persons and entities involved in civil air transportation. Laws & Regulations are needed. The compliance with the regulations has to be controlled.

Aviation is **global** requiring **International Cooperation** and International Standards

— Some international and National Aviation organizations:

ICAO · OACI · ИКАО
EASA European Aviation Safety Agency
EUROCONTROL
Federal Aviation Administration
AESA AGENCIA ESTATAL DE SEGURIDAD AÉREA
direction générale de l'Aviation civile

Aviation safety – The basics: Who is concerned



THE EUROPEAN AGENCY

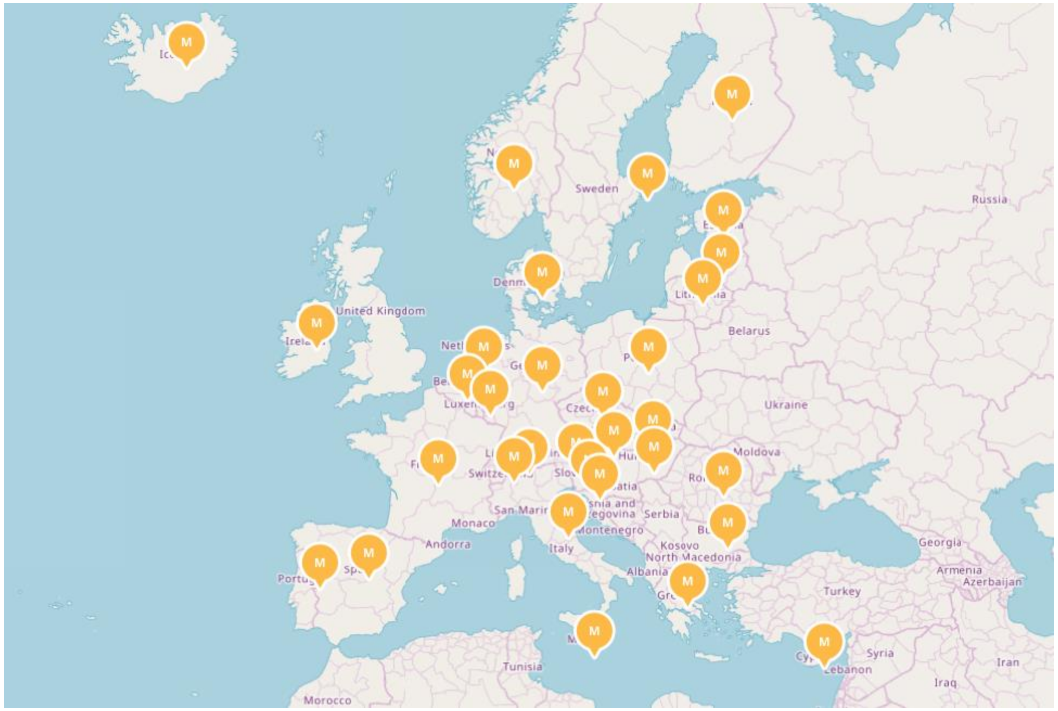
EASA builds on the experiences and cooperation of the **former** group of European aviation regulators, known as the **Joint Aviation Authorities (JAA)** which ceased its activities in July 2009.
JAA was created in **1970s** and was a group of European Aviation Authorities intending to establish a common safety standards and procedures. It **was not a regulatory** group with legal power. **Certification at this time was achieved trough each member authority.**

Nowadays Mission:

- Ensure the highest common level of safety protection for EU citizens
- Ensure the highest common level of environmental protection
- Single regulatory and certification process among Member States
- Work with other international aviation organisations & regulators

EUROPEAN AVIATION SAFETY AGENCY - EASA

EASA was established in 2002. The headquarters are sited in Cologne (Germany)



It is composed by 27 EU states + Switzerland, Norway, Iceland and Liechtenstein

The figure shows the EASA Member States

On January 1, 2021 EU aviation safety legislation no longer applies to the UK. As of that date, the UK is considered as a third country and no longer has the status of an 'EASA Member State'. EASA will no longer be acting as competent authority for the UK as State of design – this role will be taken over by the UK CAA.

CS – Certification Specification

The Agency shall issue **certification specifications**, including airworthiness codes and **acceptable means of compliance**, defining what the applicant has to demonstrate to comply with the regulations and to obtain a technical approval.

CS-22 Sailplanes and Powered Sailplanes	CS-Definitions Definitions and Abbrevations
CS-23 Normal, Utility, Aerobatic & Commuter A/C	CS-E Engines
CS-25 Large Aeroplanes	CS-ETSO European Technical Standard Orders
CS-26 Additional airworthiness specifications for operations	CS-FCD Flight Crew Data
CS-27 Small Rotorcraft	CS-FSTD (A) Aeroplane Flight Simulation Training Devices
CS-29 Large Rotorcraft	CS-FSTD (H) Helicopter Flight Simulation Training Devices
CS-31GB Gas Balloons	CS-GEN-MMEL Generic Master Minimum Equipment List
CS-31 HB Hot Air Balloons	CS-LSA Light Sport Aeroplanes
CS-31 TGB Tethered Gas Balloons	CS-MMEL Master Minimum Equipment List
CS-34 A/C Engine Emissions and Fuel Venting	CS-P Propellers
CS-36 A/C Noise	CS-SIMD Simulator Data
CS-ACNS Airborne Communications, Navigation & Surveillance	CS-VLA Very Light aeroplanes
CS-ADR-DSN Aerodromes design	CS-VLR Very Light rotorcraft
CS-APU Auxiliary Power Units	AMC-20 General Acceptable Means of Compliance
CS-AWO All Weather Operations	CS-MSCD Maintenance Certifying Staff Data
CS-CCD Cabin Crew Data	CS-OSD result from operational suitability data rulemaking

CS for Large Aeroplanes

CS of interes for Large Aeroplanes Manufacturers

Certification Specifications and Acceptable Means of Compliance for Large Aeroplanes CS – 25

- CS-25 is applicable for aircraft weight exceeding 5700 Kg defined as large aeroplanes. Contents:
  - Book 1: Structure Certification Specifications
  - Book 2: Acceptable Means of Compliance (AMC) associated to every CS

- EASA CS25 Original issue is equivalent to JAA,JAR 25 at change 16, dated 11/10/2003
  - Amdt 1: 12/12/2005
  - Amdt 2: Applicable regulation at time of initial A350 application for TC
  - Amdt 3: 19/09/2007
  - Amdt 8: 18/12/2009
  - Amdt 9: 12/08/2010 (Composites with new AMC 20-29)
  - Amdt 13: 10/06/2013
  - Amdt 14: 19/12/2013
  - Amdt 17: 16/07/2015
  - Amdt 19: 12/05/2017
  - Amdt 21: 27/03/2018
  - Amdt 22: 05/11/2018
  - Amdt 24: 10/01/2020
  - Amdt 25: 24/06/2020
  - Amdt 27: 06/12/2021
  - Amdt 28:19/12/2023

Bilateral Agreements and Working Arrangements

EASA works at facilitating the free movement of European products and services worldwide.

It **assists non-European authorities** when they certify **European products** and services. Reciprocally, **it issues European certificates for non-European** products.

The legal tools to do so are bilateral agreements and Working Arrangements.

- A **Bilateral Aviation Safety Agreement** (BASA) is signed between the **EU** (and its Member States) and **a non-EU country**. It is used when the cooperation between the two sides aims at the **mutual acceptance of certificates**. EASA supports the European Commission during the negotiation and implementation of such agreements. So far, the EU has concluded a BASA with the US, Canada, Brazil, China, Japan and UK.
- A **Working Arrangement** (WA) is usually signed between **EASA** and the **authority** of a **non-EU country**, or a regional or international organisation. It covers matters of technical nature. It is typically used to facilitate **EASA's certification tasks or the validation by a foreign authority** of EASA certificates. Unlike BASAs, WAs **do not** allow for the **mutual recognition of certificates**. EASA directly negotiates and concludes such arrangements.

US National Aviation Authority - The FAA

The Federal Aviation Administration (**FAA**) was established in 1958, when the Civil Aeronautics Authority's functions were transferred to a new **independent Federal Aviation Agency responsible for civil aviation safety**. Becoming later in Federal Aviation Administration (FAA) as an organization within Department of Transportation (DOT).
The headquarters are in Washington D.C.

Objective:

- Set the rules of **all** aspects of **civil aviation in the USA**.
- This rules are **binding** rules unlike EASA

Tasks:

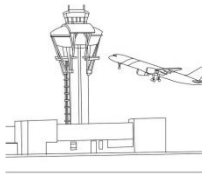
- Issue the Federal Aviation Regulations (FARs)
- Implementation of regulation control
- Enforce compliance
- Rulemaking and changes to rules involving the public

FARs – Federal Airworthiness Regulations

FARs are published as Title 14 “Aeronautics & Space” of the Code of Federal Regulations (Chapter1). The official nomenclature is “14 CFR Part xx” which is the “Chapter 14 of the Code of Federal Regulations – Part xx”. However the common named used is FARxx. The FARs affecting Aircrafts are:

- Procedural Requirements:
  - » FAR 21: Certification Procedures for Products and Parts
- Airworthiness Standards:

  - » FAR 25: Transport Category Aircraft
  - » FAR 26: Continued Airworthiness and Safety Improvements
- Operational Requirements



  - » FAR 91 General Operating and Flight Rules
  - » FAR 121: Domestic, Flag and Supplemental Operations
  - » FAR 129: Foreign Air Carriers and Foreign Operators of US registered Aircraft engaged in Common Carriage

Regulations correspondence between EASA and FAA

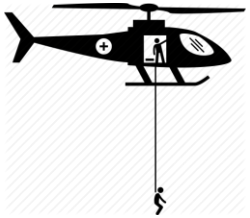
	EASA	FAA
Certification Procedures	Part 21	FAR 21
Large Aeroplanes Design Requirements	CS-25	FAR 25
Airworthiness Directives	Part 21	FAR 39
Maintenance	Part M	FAR 43
Repairs	Part 21 Subpart M	FAR 43
Maintenance and Licenses	Part 66	FAR 65
Maintenance Organization	Part 145	FAR 145
Maintenance Training Schools	Part 147	FAR 147
Operations	IR OPS 965/2012	FAR 121

AIRWORTHINESS IN MILITARY AVIATION

AIRWORTHINESS IN MILITARY AVIATION

- EASA has not the exclusive competences in civil aviation in Europe. Some of the competences are retained by the national civil aviation authorities, referred to in EASA regulation as Member States **Competent Authorities**, but under the EASA supervision.
- In addition EASA **is not competent** either for products, parts, appliances, personnel and organizations engaged in military, fire fighting, customs, police, or similar services.





- The Member States shall undertake to ensure that such services have due regard as far as practicable to the objectives of this Regulation.
- Therefore, although EASA is the civil aviation authority in accordance with a law approved by the European Commission and the European Parliament, but it is not the only aviation authority.

AIRWORTHINESS IN MILITARY AVIATION

• Main differences in airworthiness between civil and military aviation is the fact that:

- No general international conventions like the Chicago one exists in military aviation.
- The **nations** are fully responsible of the **airworthiness** of aircraft operating under their **military register**. This includes initial certification and continuous airworthiness.
- The higher complexity of military aircraft types and **missions** has direct impact in **airworthiness**.
- Certification and qualification aspects can not be easily managed separately.

AIRWORTHINESS IN MILITARY AVIATION

- As a consequence:
  - There are **no uniform criteria** for the certification processes.
  - Only a few countries have a relevant airworthiness frame.
  - Only in multi-national projects, agreements are reached.
  - Each project is managed individually.
  - Individual tailoring of certification basis is the common practice characteristic
  - Technical regulations need **quick evolution** as military technology grows.

EDA

- European Defence Agency (EDA) was formed 2004 in with the primary role to foster European Defence cooperation.

- All EU member states take part in EDA except Denmark.

- The EU Defence Ministers agreed in November 2008 the formation of the Military Airworthiness Authorities (MAWA) Forum with the aim to harmonise the European military airworthiness regulations of Member States.

EUROPEAN DEFENCE AGENCY

MAWA

- The MAWA Forum comprises representatives from the Military Airworthiness Authorities of Member States and industry representatives and is chaired by the EDA (European Defense Agency).

- A single European Military Airworthiness Organisation (JAA Model) owning a suite of European Military Airworthiness Requirements (**EMAR**) used by all participating Member States to govern peacetime European Military Airworthiness activities facilitated by Mutual Recognition, consistent implementation and Standard Industry Arrangements including Obligations and Privileges.

- Each nation in the EU operates an independent military aviation regulatory system and is individually responsible for the regulation of its own military aircraft.

Unfortunately, for the airworthiness of military aircraft, and in order to protect the sovereignty,

There is not a consensus in the pMS intent to support the creation of an European authority similar to EASA.

EUROPEAN DEFENCE AGENCY

EMAR

- MAWA does not have the authority to impose airworthiness regulations upon nations, who retain their sovereignty for managing military airworthiness.
- Unfortunately, in most cases, **EMAR's are not directly transferred to national law**.
- The decision of whether to implement the harmonised EMAR's, and the timescales in which this will be done, remains the responsibility of the participating Member States (pMS).
- This process of transfer to national law means in many cases a **customization** of the EMAR's, that leads to a lack of harmonization between the laws in the different pMS, normally to a further deviation from the EASA regulations and additional difficulties in the mutual recognition of the approvals by the pMS.

EMAR

- In the development of EMAR’s, and in particular **EMAR Part 21**, the basic principle has been to maintain, as far as possible, the content, formatting and paragraph numbering of EASA Part 21, to facilitate the comparison of the rules and to harmonise the processes in the civil and military fields.

- One of the consequences of the implementation of the EMAR’s is the change of the classical military processes towards the principles of the civil regulations that rely on the **Industry to assume privileges**, reducing the involvement of the airworthiness authorities, implementing more efficient processes and reducing timescales.

SAFETY AND MILITARY OPERATIONAL CAPABILITY

•In military aviation, an appropriate **balance** between the operational **requirements** and the adequate level of **safety** shall exist.

•In military aircrafts, the mission requirements are essential. This makes very critical the adequate selection of the safety/airworthiness level for each air-system and the missions to be fulfilled.

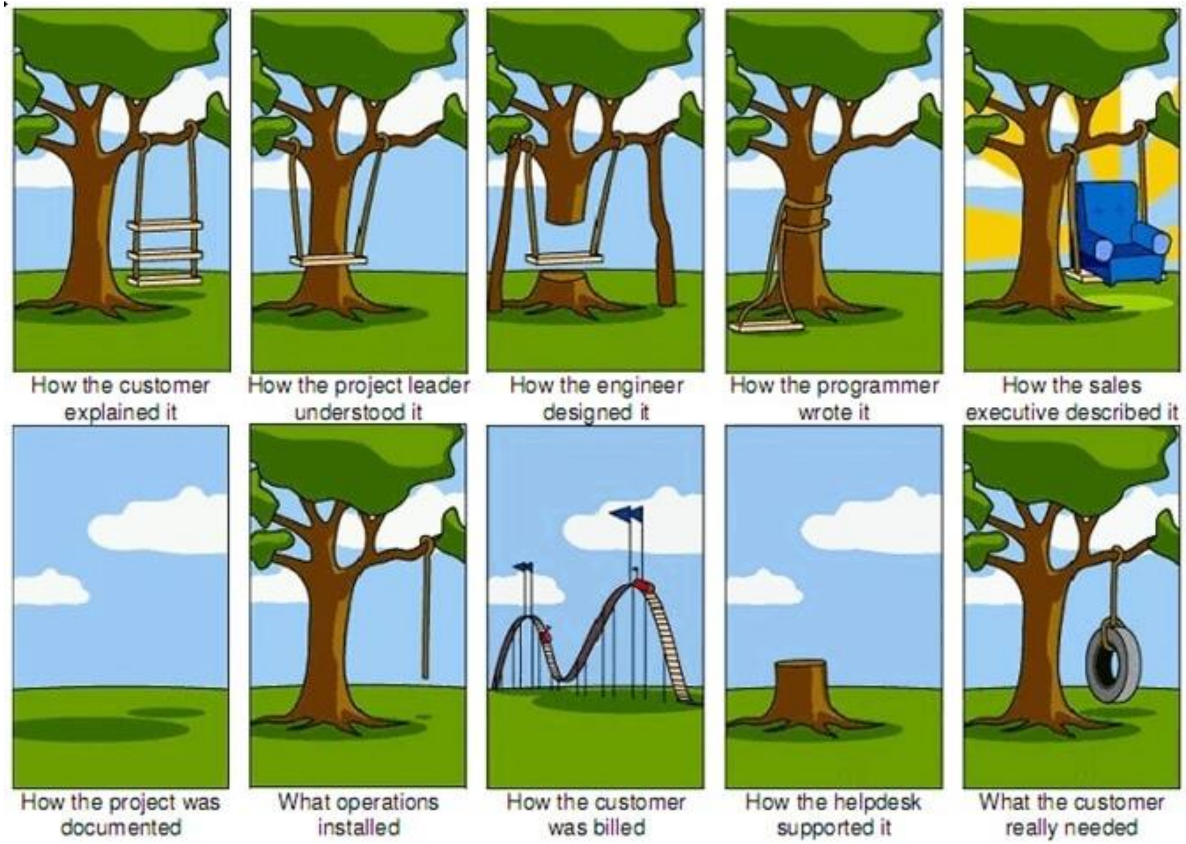
- A too high or too low safety standard can make the mission impractical.
- If the **mission** is **impractical** there is **<u>no aircraft</u>**

QUALIFICATION

Qualification

Qualification is a term used in military aviation as:

The demonstrated capability of an air-system to perform the tasks required in its **contractual specification** in a satisfactory manner, in front of the Authorities designated by the **customer**.



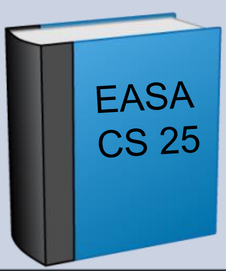
Qualification

Certification ensures that an aircraft is "**fit for flight**". This means that the appropriate airworthiness level has been demonstrated.

Qualification ensures that an aircraft is "**fit for purpose**". This means that the required capabilities of the aircraft as military platform have been demonstrated.

In some military aeronautical projects, the certification is part of the qualification process e.g. demonstration of the airworthiness requirements of the technical specification is part of the overall demonstration process.

Qualification

<table>
<tr><th>Certification</th><th>Qualification</th></tr>
<tr><td colspan="2">It is the process of demonstration that the design of the product complies with</td></tr>
<tr><td>The applicable airworthiness requirements</td><td>The Technical Specification Requirements (contract)</td></tr>
<tr><td>in front of Airworthiness Authorities

</td><td>in front of customer
(or the authority stated in the contract)

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Thank you !